

1. Use algebra to determine the values of x for which

$$x(x-1) > \frac{x-1}{x}$$

giving your answer in set notation.

(6)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2. The variables x and y satisfy the differential equation

$$\frac{d^2y}{dx^2} + 15\frac{dy}{dx} - 3y^2 = 2x$$

where $y = 1$ at $x = 0$ and where $y = 2$ at $x = 0.1$

Use the approximations

$$\left(\frac{d^2y}{dx^2}\right)_n \approx \frac{(y_{n+1} - 2y_n + y_{n-1}))}{h^2} \text{ and } \left(\frac{dy}{dx}\right)_n \approx \frac{(y_{n+1} - y_{n-1}))}{2h}$$

with $h = 0.1$ to find an estimate for the value of y when $x = 0.3$

(6)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 6 marks)



3. On a particular day, the depth of water in a river estuary at a specific location is modelled by the equation

$$D = 2 \sin\left(\frac{x}{3}\right) + 3 \cos\left(\frac{x}{3}\right) + 6 \quad 0 \leq x \leq 7\pi \quad (\text{I})$$

where the depth of water is D metres at time x hours after midnight on that day.

- (a) Write down the depth of water at midnight, according to the model.

(1)

Using the substitution $t = \tan\left(\frac{x}{6}\right)$

- (b) show that equation (I) can be re-written as

$$D = \frac{3t^2 + 4t + 9}{1 + t^2} \quad (3)$$

- (c) Hence determine, according to the model, the time after midnight when the depth of water is 5 metres for the first time. Give your answer to the nearest minute.

(5)



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Question 3 continued

Lined area for writing the answer to Question 3 continued.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4. With respect to a fixed origin O , the points A , B and C have position vectors given by

$$\overrightarrow{OA} = 18\mathbf{i} - 14\mathbf{j} - 2\mathbf{k} \quad \overrightarrow{OB} = -7\mathbf{i} - 5\mathbf{j} + 3\mathbf{k} \quad \overrightarrow{OC} = -2\mathbf{i} - 9\mathbf{j} - 6\mathbf{k}$$

The points O, A, B and C form the vertices of a tetrahedron.

- (a) Show that the area of the triangular face ABC of the tetrahedron is 108 to 3 significant figures.

(3)

- (b) Find the volume of the tetrahedron.

(4)

An oak wood block is made in the shape of the tetrahedron, with centimetres taken for the units.

The density of oak is 0.85 g cm^{-3}

- (c) Determine the mass of the block, giving your answer in kg.

(2)



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Question 4 continued

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Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 9 marks)



5. The point $P(ap^2, 2ap)$, where a is a positive constant, lies on the parabola with equation

$$y^2 = 4ax$$

The normal to the parabola at P meets the parabola again at the point $Q(aq^2, 2aq)$

- (a) Show that

$$q = \frac{-p^2 - 2}{p} \quad (5)$$

- (b) Hence show that

$$PQ^2 = \frac{ka^2}{p^4}(p^2 + 1)^n$$

where k and n are integers to be determined.

(5)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 10 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 1 IS 40 MARKS

